

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

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Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit I | Session 4 | SLO 1: Multiple Programming Paradigms (Code: 1041)

Q: 1. What is a programming paradigm?

Ans: Definition: A programming paradigm is a fundamental classification, overarching style, or architectural methodology of computer programming that defines how computation is structured and organized.

Q: 2. Name any three common programming paradigms.

Ans: 1. **Imperative / Procedural Paradigm** (e.g., C)
2. **Object-Oriented Paradigm (OOP)** (e.g., Java, C++)
3. **Functional Paradigm** (e.g., Haskell, Scala, Lisp)

Q: 3. Why is it useful for a language to support multiple paradigms?

Ans: Utility: Multi-paradigm languages (like modern Java, Python, and C++) allow developers to apply the optimal approach for each distinct task. For example, using declarative functional streams for high-speed data transformations while using object-oriented classes for complex business domain modeling.

Q: 4. Which paradigm focuses on real-world entities like objects and classes?

Ans: Answer: Object-Oriented Programming (OOP) Paradigm.

Q: 5. Which paradigm emphasizes step-by-step instructions to change program state?

Ans: Answer: Imperative Programming Paradigm.